

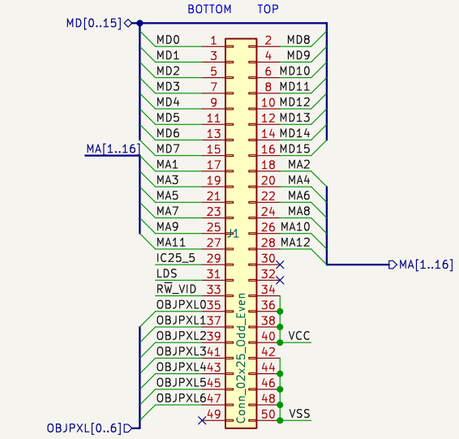


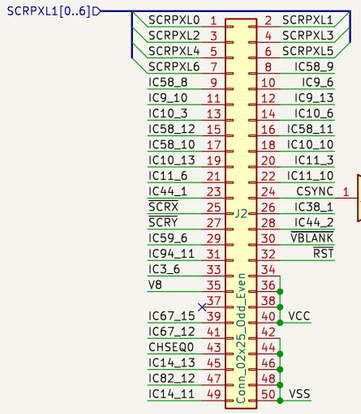
Leo Tejada
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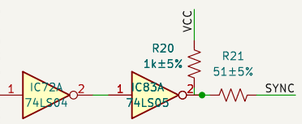
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File: wwfsstar.kicad_sch

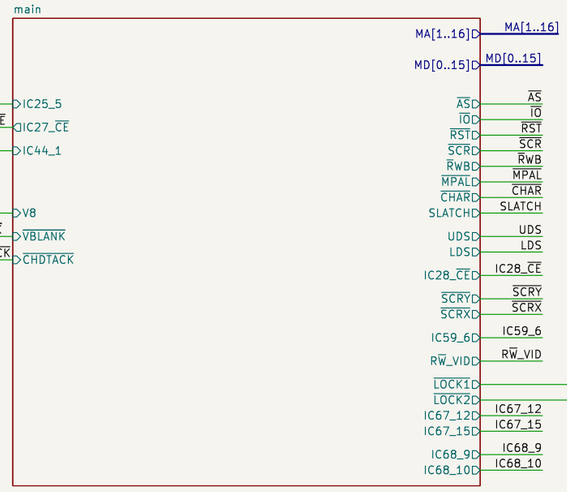
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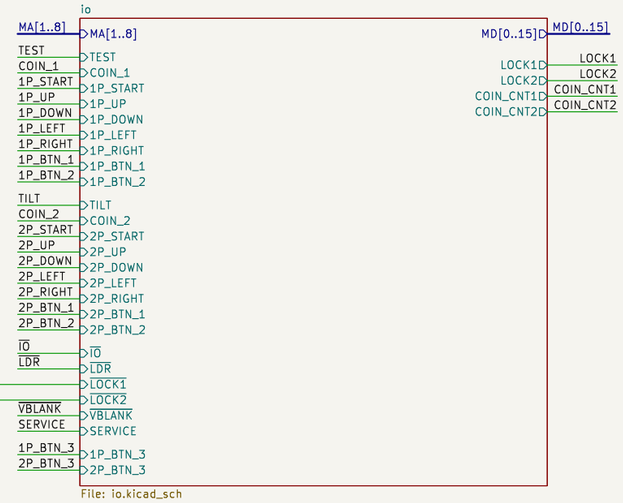
Size: A3	Date:	**Rev:**
KiCad E.D.A. 8.0.5		Id: 1/20

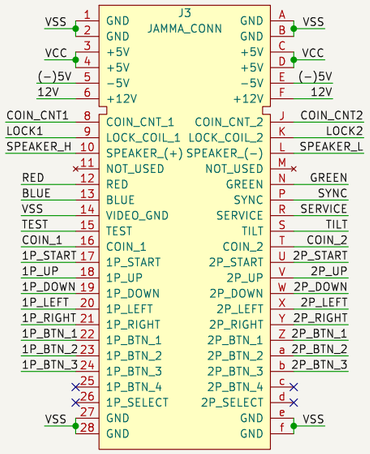

BOTTOM TOP
MD[0..15]
MD0 1
MD1 3
MD2 5
MD3 7
MD4 9
MD5 11
MD6 13
MD7 15
MA1 17
MA3 19
MA5 21
MA7 23
MA9 25
MA11 27
IC25_5 29
LDS 31
R/W_VID 33
OBJPXL0 35
OBJPXL1 37
OBJPXL2 39
OBJPXL3 41
OBJPXL4 43
OBJPXL5 45
OBJPXL6 47
49
2 MD8
4 MD9
6 MD10
8 MD11
10 MD12
12 MD13
14 MD14
16 MD15
18 MA2
20 MA4
22 MA6
24 MA8
26 MA10
28 MA12
30
32
34
36
38
40 VCC
42
44
46
48
50 VSS
J1
Conn_02x25_Odd_Even
MA[1..16]
MA[1..16]
OBJPXL[0..6]

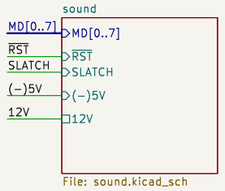

SCRPXL1[0..6]
SCRPXL0 1
SCRPXL2 3
SCRPXL4 5
SCRPXL6 7
IC58_8 9
IC9_10 11
IC10_3 13
IC58_12 15
IC58_10 17
IC10_13 19
IC11_6 21
IC44_1 23
SCRX 25
SCRY 27
IC59_6 29
IC94_11 31
IC3_6 33
V8 35
37
IC67_15 39
IC67_12 41
CHSEQ0 43
IC14_13 45
IC82_12 47
IC14_11 49
2 SCRPXL1
4 SCRPXL3
6 SCRPXL5
8 IC58_9
10 IC9_6
12 IC9_13
14 IC10_6
16 IC58_11
18 IC10_10
20 IC11_3
22 IC11_10
24 CSYNC
26 IC38_1
28 IC44_2
30 VBLANK
32 RST
34
36
38
40 VCC
42
44
46
48
50 VSS
J2
Conn_02x25_Odd_Even

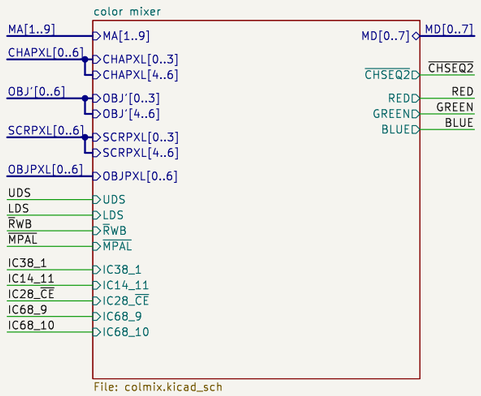

IC72A
74LS04
IC83A
74LS05
R20
1k±5%
R21
51±5%
VCC
SYNC

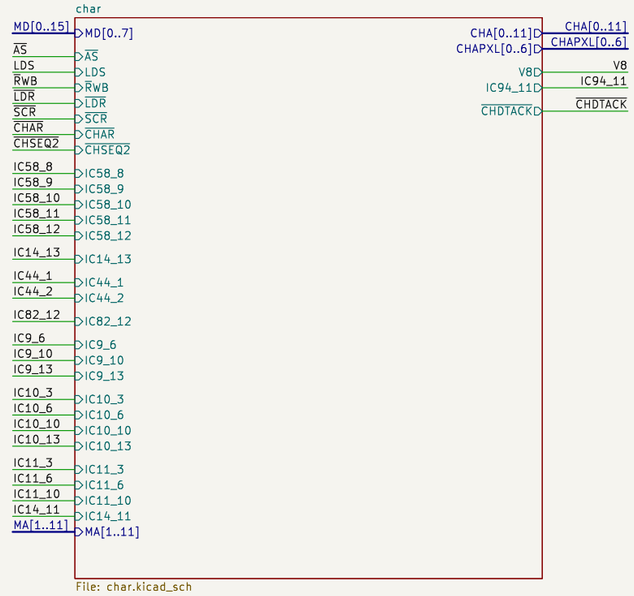

main
IC25_5
IC27_CE
IC44_1
V8
VBLANK
CHDTACK
MA[1..16]
MD[0..15]
AS
IO
RST
SCR
R/W
MPAL
CHAR
SLATCH
UDS
LDS
IC28_CE
SCRY
SCRX
IC59_6
R/W_VID
LOCK1
LOCK2
IC67_12
IC67_15
IC68_9
IC68_10
MA[1..16]
MD[0..15]
File: main.kicad_sch

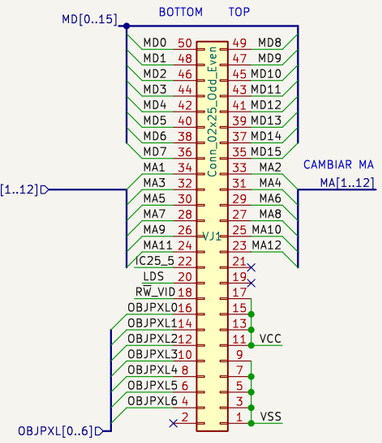

io
MA[1..8]
TEST
COIN_1
1P_START
1P_UP
1P_DOWN
1P_LEFT
1P_RIGHT
1P_BTN_1
1P_BTN_2
TILT
COIN_2
2P_START
2P_UP
2P_DOWN
2P_LEFT
2P_RIGHT
2P_BTN_1
2P_BTN_2
IO
LDR
LOCK1
LOCK2
VBLANK
SERVICE
1P_BTN_3
2P_BTN_3
MD[0..15]
LOCK1
LOCK2
COIN_CNT1
COIN_CNT2
File: io.kicad_sch

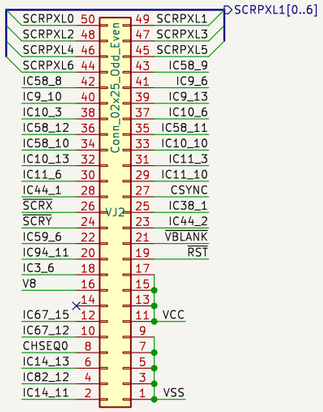

J3
JAMMA_CONN
VSS 1
VSS 2
VCC 3
VCC 4
(−)5V 5
12V 6
COIN_CNT1 8
LOCK1 9
SPEAKER_H 10
11
RED 12
BLUE 13
VSS 14
TEST 15
COIN_1 16
1P_START 17
1P_UP 18
1P_DOWN 19
1P_LEFT 20
1P_RIGHT 21
1P_BTN_1 22
1P_BTN_2 23
1P_BTN_3 24
25
26
VSS 27
28
GND GND
+5V +5V
+5V +5V
−5V −5V
+12V +12V
COIN_CNT_1 COIN_CNT_2
LOCK_COIL_1 LOCK_COIL_2
SPEAKER_(+) SPEAKER_(−)
NOT_USED NOT_USED
RED GREEN
BLUE SYNC
VIDEO_GND SERVICE
TEST TILT
COIN_1 COIN_2
1P_START 2P_START
1P_UP 2P_UP
1P_DOWN 2P_DOWN
1P_LEFT 2P_LEFT
1P_RIGHT 2P_RIGHT
1P_BTN_1 2P_BTN_1
1P_BTN_2 2P_BTN_2
1P_BTN_3 2P_BTN_3
1P_BTN_4 2P_BTN_4
1P_SELECT 2P_SELECT
GND GND
GND GND
A VSS
B VSS
C VCC
D VCC
E (−)5V
F 12V
J COIN_CNT2
K LOCK2
L SPEAKER_L
M
N GREEN
P SYNC
R SERVICE
S TILT
T COIN_2
U 2P_START
V 2P_UP
W 2P_DOWN
X 2P_LEFT
Y 2P_RIGHT
Z 2P_BTN_1
a 2P_BTN_2
b 2P_BTN_3
c
d
e
f VSS

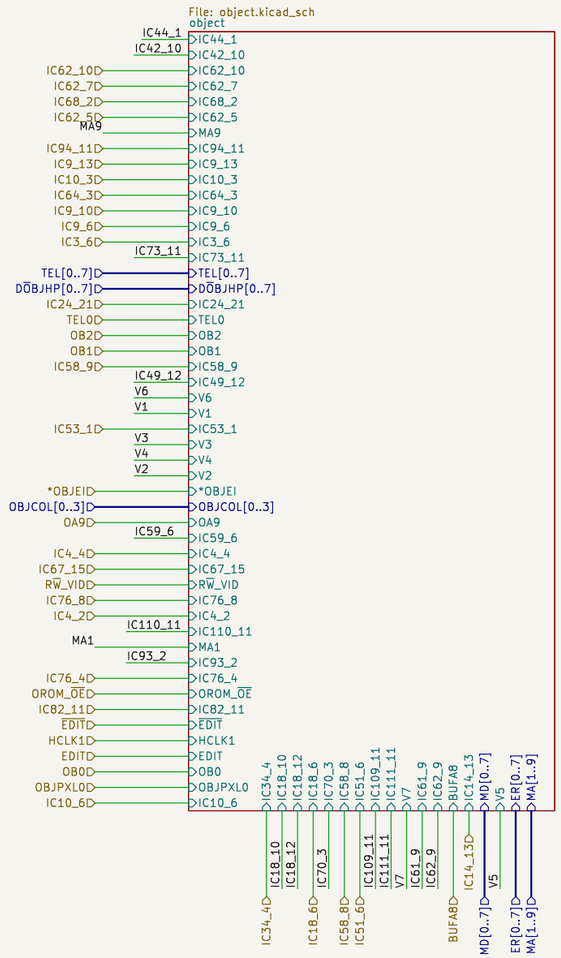

sound
MD[0..7]
RST
SLATCH
(−)5V
12V
MD[0..7]
RST
SLATCH
(−)5V
12V
File: sound.kicad_sch

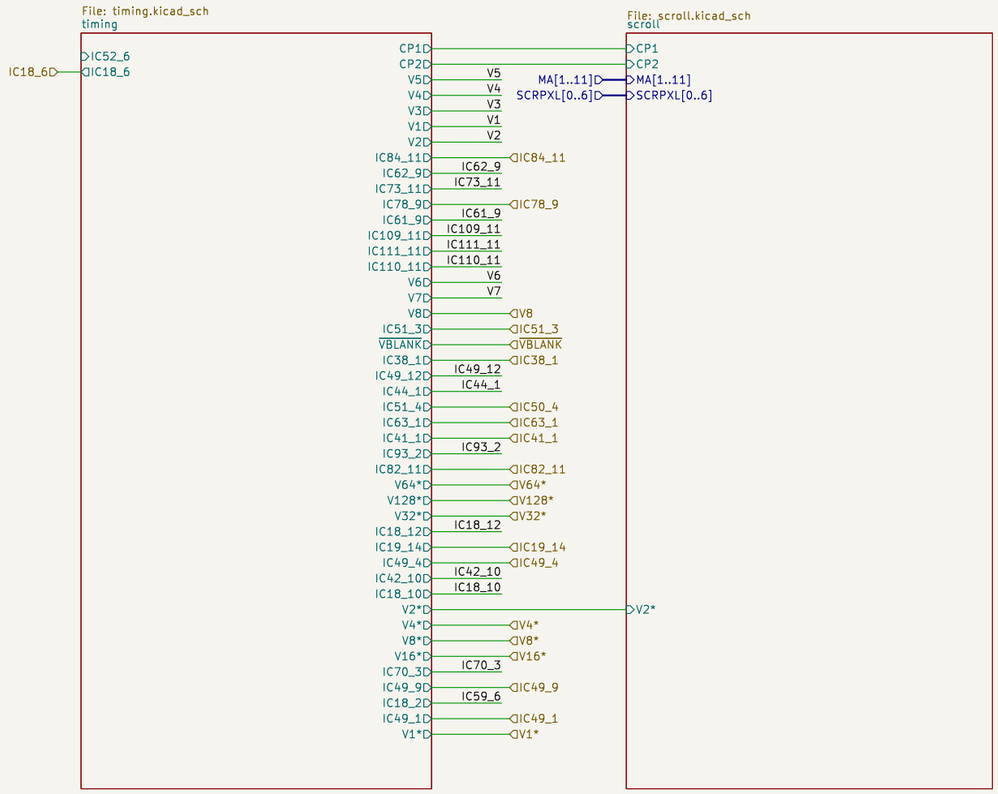

color mixer
MA[1..9]
CHAPXL[0..6]
OBJ'[0..6]
SCRPXL[0..6]
OBJPXL[0..6]
UDS
LDS
R/W
MPAL
IC38_1
IC14_11
IC28_CE
IC68_9
IC68_10
MD[0..7]
CHSEQ2
RED
GREEN
BLUE
MA[1..9]
CHAPXL[0..3]
CHAPXL[4..6]
OBJ'[0..3]
OBJ'[4..6]
SCRPXL[0..3]
SCRPXL[4..6]
OBJPXL[0..6]
UDS
LDS
R/W
MPAL
IC38_1
IC14_11
IC28_CE
IC68_9
IC68_10
File: colmix.kicad_sch


char
MD[0..15]
AS
LDS
R/W
LDR
SCR
CHAR
CHSEQ2
IC58_8
IC58_9
IC58_10
IC58_11
IC58_12
IC14_13
IC44_1
IC44_2
IC82_12
IC9_6
IC9_10
IC9_13
IC10_3
IC10_6
IC10_10
IC10_13
IC11_3
IC11_6
IC11_10
IC14_11
MA[1..11]
MD[0..7]
AS
LDS
R/W
LDR
SCR
CHAR
CHSEQ2
IC58_8
IC58_9
IC58_10
IC58_11
IC58_12
IC14_13
IC44_1
IC44_2
IC82_12
IC9_6
IC9_10
IC9_13
IC10_3
IC10_6
IC10_10
IC10_13
IC11_3
IC11_6
IC11_10
IC14_11
MA[1..11]
CHA[0..11]
CHAPXL[0..6]
V8
IC94_11
CHDTACK
CHA[0..11]
CHAPXL[0..6]
V8
IC94_11
CHDTACK
File: char.kicad_sch

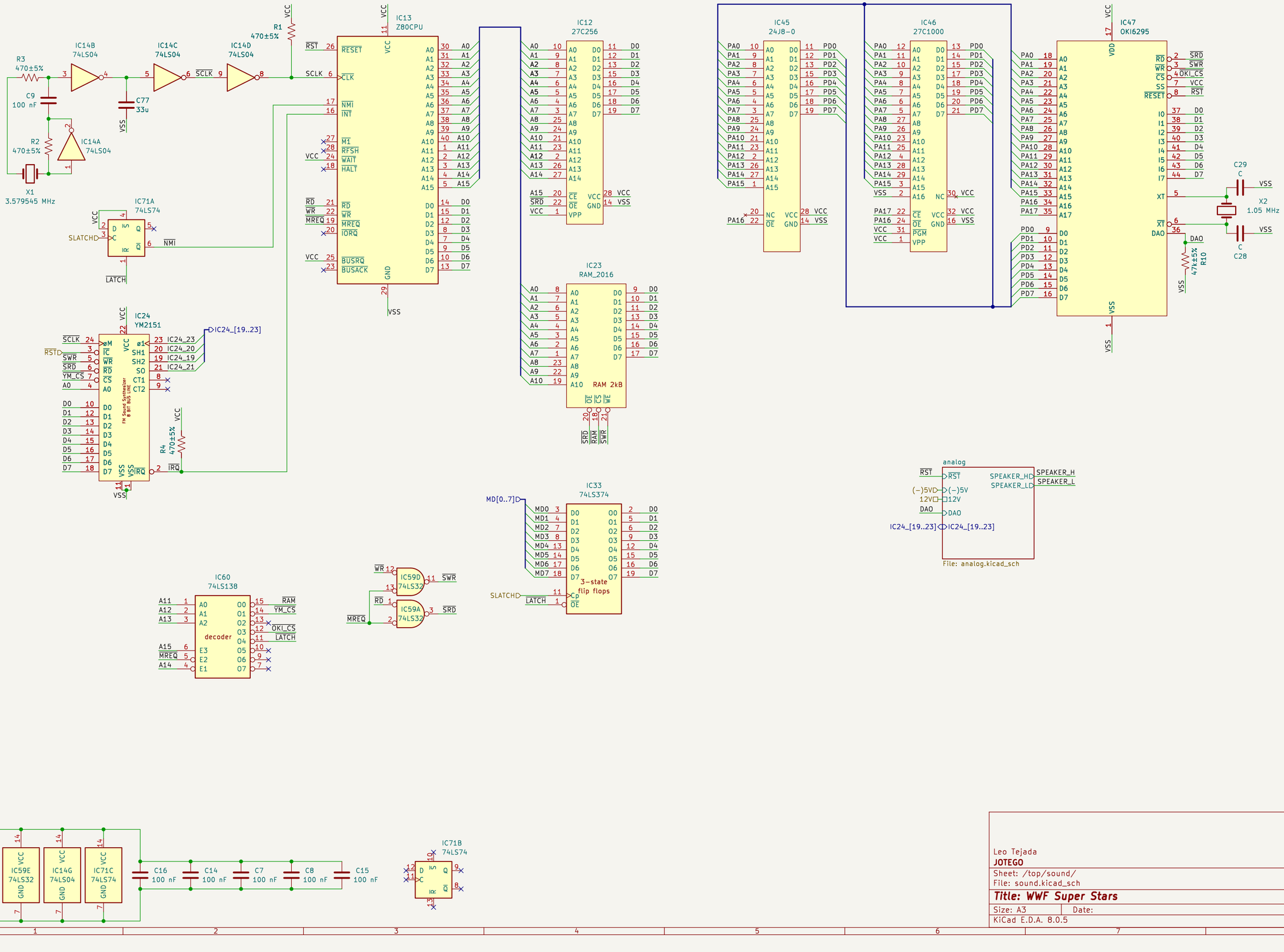


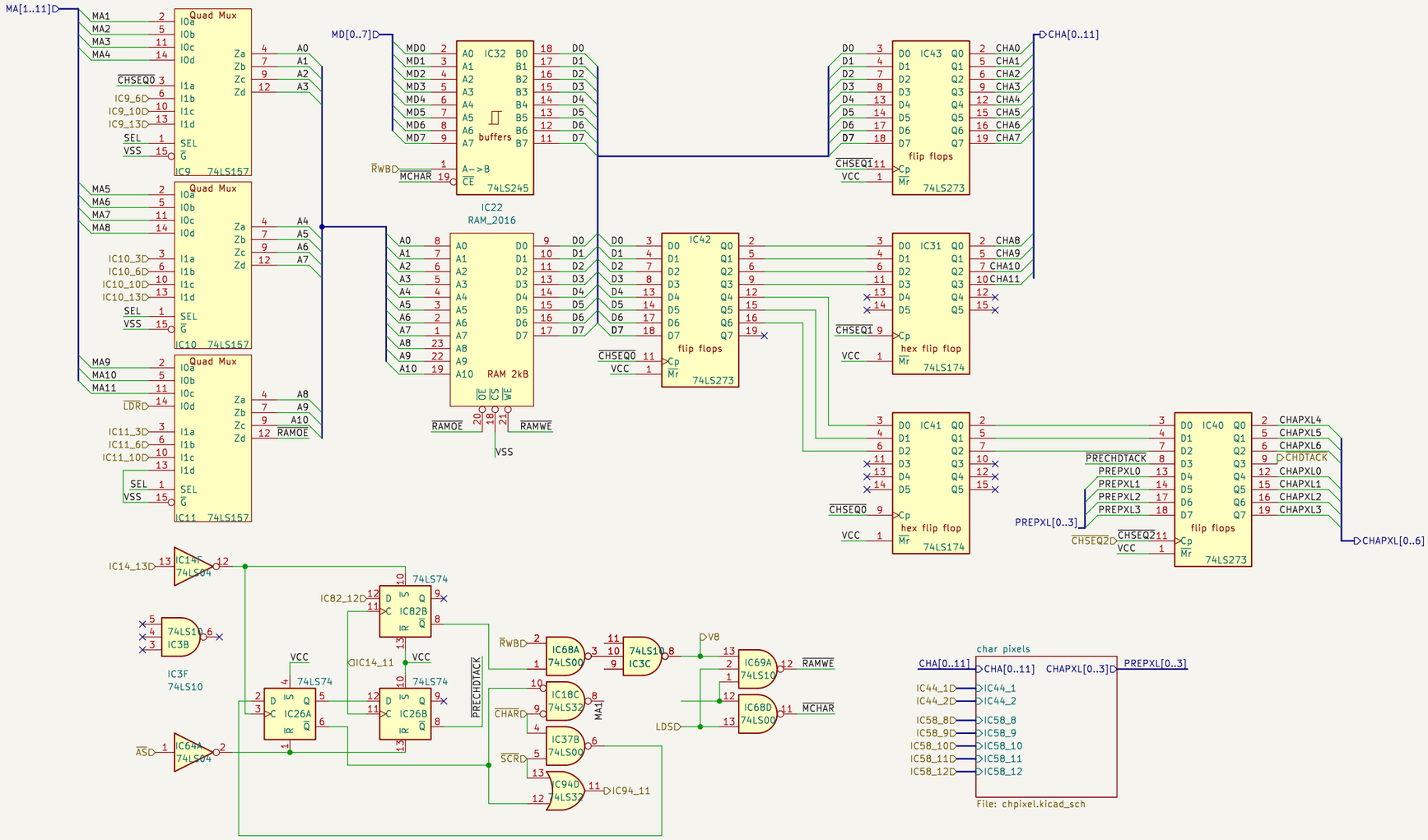




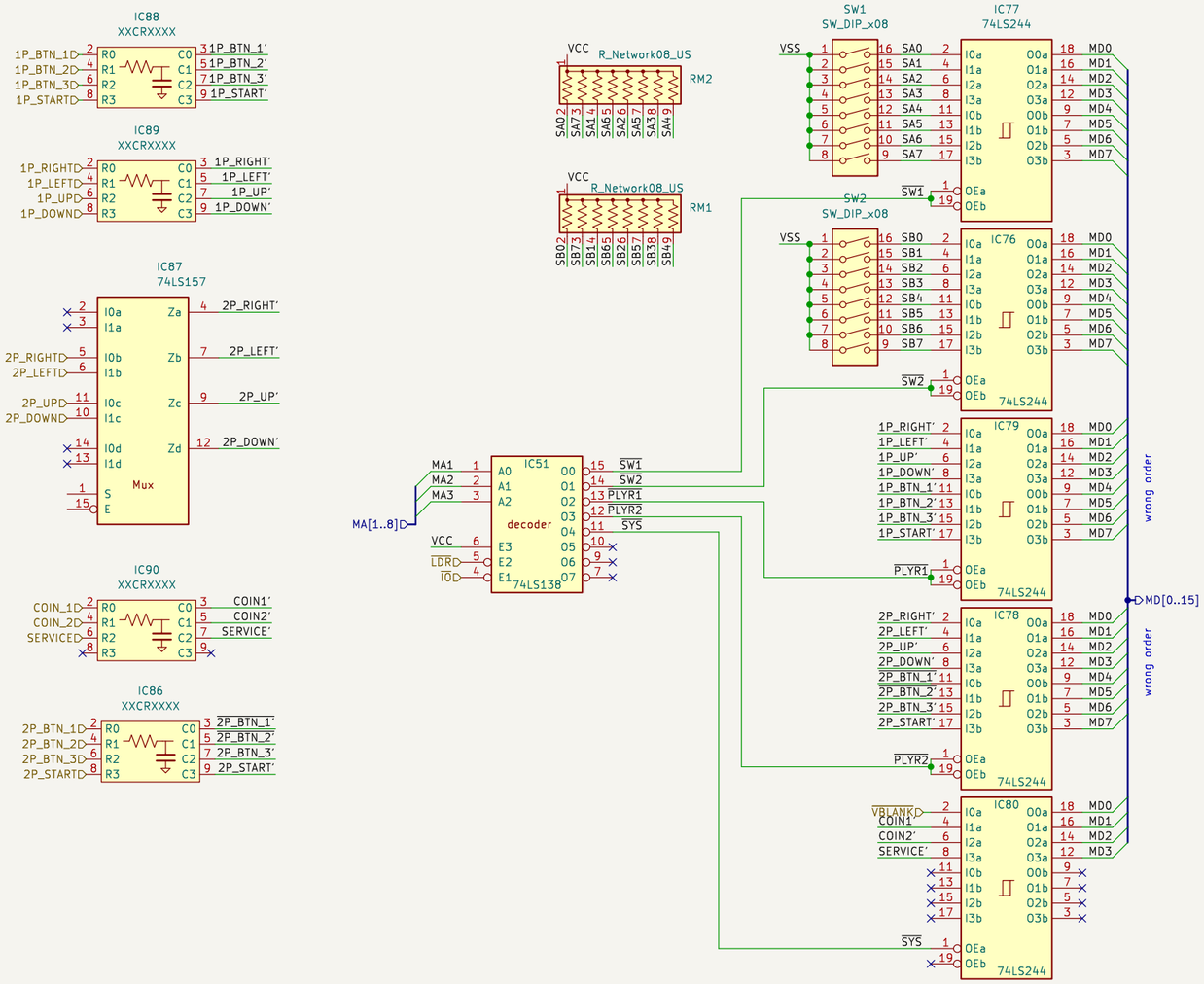


JOTEGO		
Sheet: /bottom/		
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Size: A3	Date:	**Rev:**
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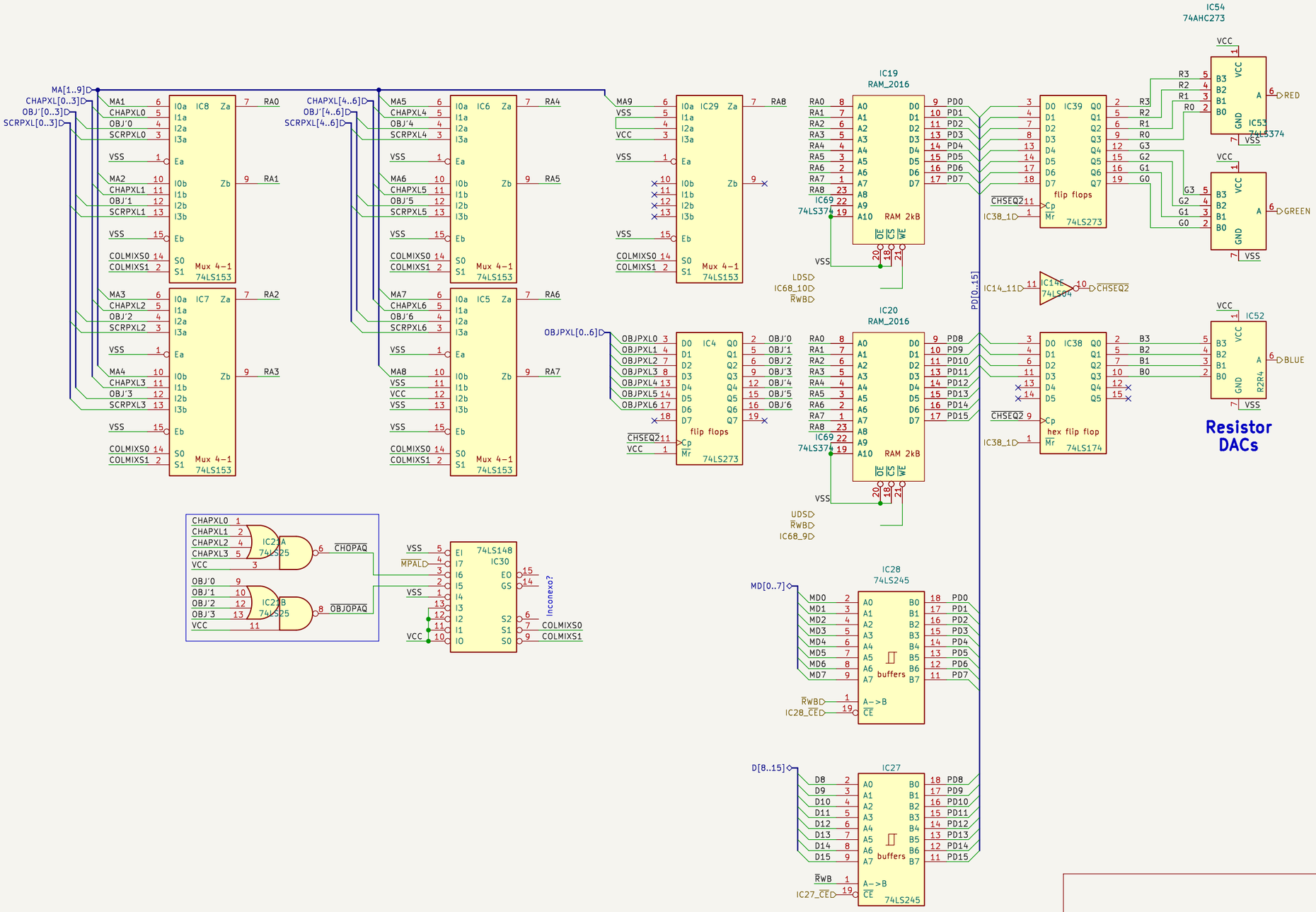

IC13
Z80CPU
IC12
27C256
IC45
24J8-0
IC46
27C1000
IC47
OKI6295
IC23
RAM_2016
IC24
YM2151
IC71A
74LS74
IC14B
74LS04
IC14C
74LS04
IC14D
74LS04
IC14A
74LS04
R1
470±5%
R2
470±5%
R3
470±5%
R4
470±5%
C9
100 nF
C77
33u
X1
3.579545 MHz
X2
1.05 MHz
C29
C28
R10
47k±5%
IC33
74LS374
3-state
flip flops
IC60
74LS138
decoder
IC59D
74LS32
IC59A
74LS32
analog
File: analog.kicad_sch
IC84G
74LS14
IC59E
74LS32
IC14G
74LS04
IC71C
74LS74
C16
100 nF
C14
100 nF
C7
100 nF
C8
100 nF
C15
100 nF
IC71B
74LS74
RAM 2kB
Leo Tejada
JOTEGO
Sheet: /top/sound/
File: sound.kicad_sch
Title: WWF Super Stars
Size: A3
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MA[1..11]
MD[0..7]
CHA[0..11]
Quad Mux
IC9 74LS157
IC10 74LS157
IC11 74LS157
IC32 buffers
74LS245
IC22
RAM_2016
RAM 2kB
IC42
IC43
IC41
IC40
flip flops
74LS273
hex flip flop
74LS174
RAMOE
RAMWE
VSS
VCC
CHSEQ0
CHSEQ1
CHSEQ2
RWB
MCHAR
LDR0
SEL
VSS
PRECHDTACK
PREPXL[0..3]
CHAPXL[0..6]
CHDTACK
IC14F
74LS04
IC82A
IC82B
IC26A
IC26B
74LS74
IC3B
74LS10
IC3F
IC3C
IC68A
IC68D
74LS00
IC18C
74LS32
IC37B
IC94D
IC69A
AS
CHAR
SCR
LDS
V8
MA1
IC94_11
char pixels
CHA[0..11]
CHAPXL[0..3]
PREPXL[0..3]
File: chpixel.kicad_sch
IC69D
IC68E
IC37E
IC18E
IC94E
IC26C
IC36B
IC25D
IC3G
IC72G
IC82C
IC3D
JOTEGO
Sheet: /top/char/
File: char.kicad_sch
Title:
Size: A3
Date:
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IC63
68000D
CLK
10 MHz
IPL0
IPL1
IPL2
BGACK
BG
BR
FC0
FC1
FC2
VMA
E
VPA
BERR
DTACK
HALT
RESET
AS
UDS
LDS
R/W
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
D12
D13
D14
D15
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
A13
A14
A15
A16
A17
A18
A19
A20
A21
A22
A23
VCC
VSS
74LS74
IC66B
IC66A
IC65
74LS148
V8D
every
16 lines
IRQ5_CLR
VBLANK
IRQ6_CLR
IC25C
74LS10
IC18A
74LS32
R23
47±5% 10k±5%
R22
R13
47±1%
R12
47±1%
C84A
74LS14
C83B
74LS05
C83A
74LS05
power-on reset
C62
C_Polarized_US
C84B
74LS14
C84C
74LS14
RST
RFIX1
47±1%
33u
CFIX1
piggyback
soldered
X3
20.000MHz
IC82A
74LS74
CLK
25V
C61
IC75
IC62
IC2
IC93
IC49
74LS244
74LS245
buffers
A->B
MA[1..16]
MD[0..15]
IC17C
74LS10
IC17B
74LS10
ROM
LDS
RAM
UDS
RWB
CE
OEa
OEb
IC35
27C1000
IC34
27C1000
IC16
RAM_2016
IC15
RAM_2016
8kB
RAM
CS2
CS1
OE
WR
RAM
LDR
UDR
LDW
PGM
VPP
CE
OE
NC
IC48
IC50
decoder
74LS138
CHAR
SCR
OBJ
PAL
IO
SCRX
SCRY
SLATCH
IC67
hex flip flop
74LS174
LOCK2
LOCK1
IC67_12
IC67_15
IC64H
74LS04
IC64D
IC64E
IC64C
IC64B
IC1A
74LS00
IC37D
IC37A
IC37C
IC36A
74LS30
IC59B
74LS32
IC59C
74LS32
IC59_6
IC25B
74LS10
IC25_5
IC72B
IC68C
IC68B
IC68_10
IC68_9
IC28_CE
IC27_CE
IC1D
MPAL
RW_VID
CHDTACK
IC3A
74LS10
IC18D
74LS32
IC17A
IC1B
IC1C
IC18B
IC64F
IC83C
IC72D
IC72F
IC64G
IC1E
IC17D
IC66C
IC83G
IC21C
74LS25
JOTEGO
Sheet: /top/main/
File: main.kicad_sch
Title:
Size: A3
Date:
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KiCad E.D.A. 8.0.5
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JOTEGO	
Sheet: /top/io/	
File: io.kicad_sch	
Title: CABINET I/O	
Size: A3	Date:
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	Rev:

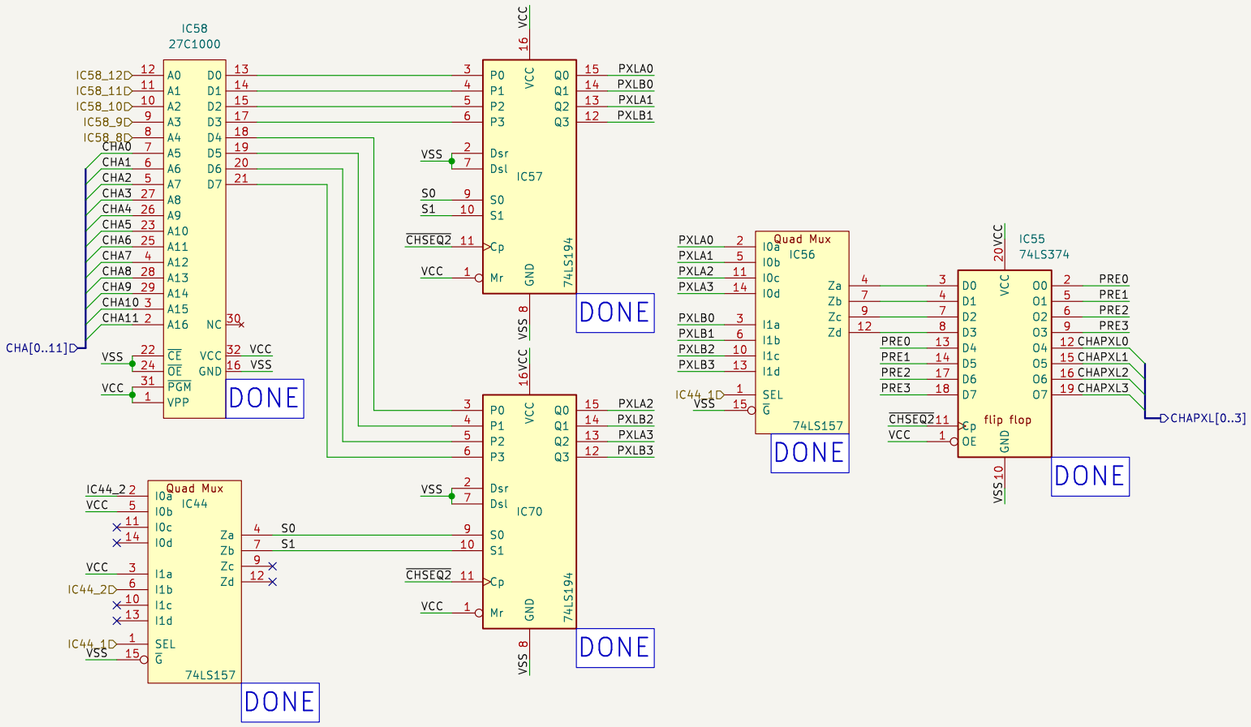


JOTEGO

Sheet: /top/color mixer/
File: colmix.kicad_sch

Title:

Size: A3	Date:	**Rev:**
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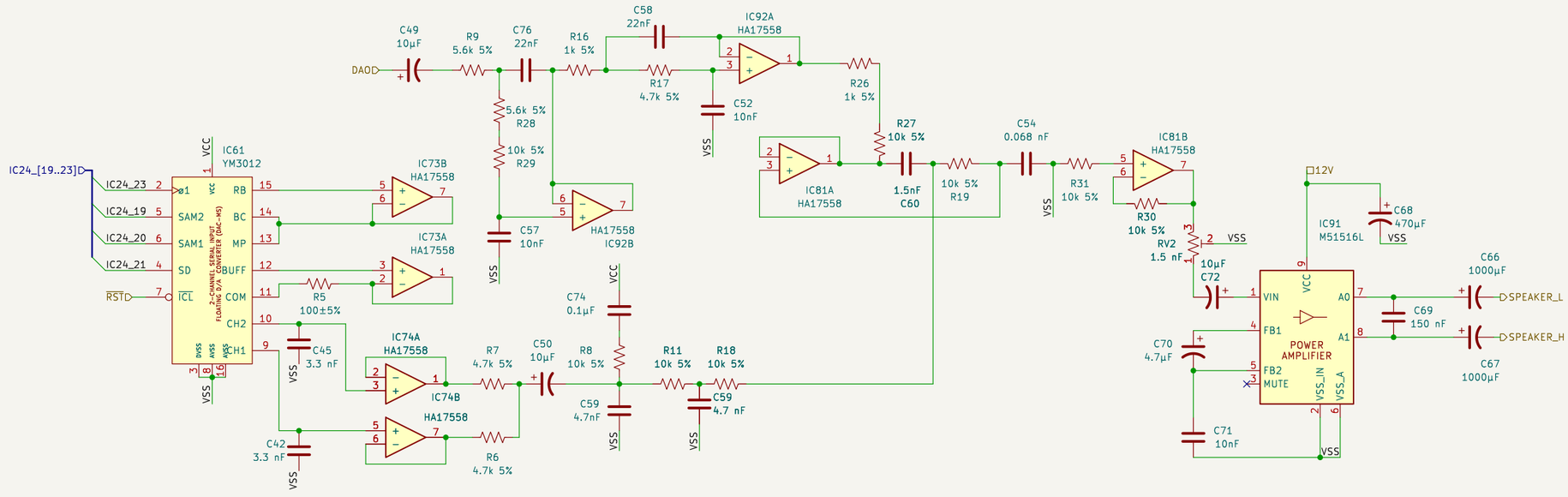


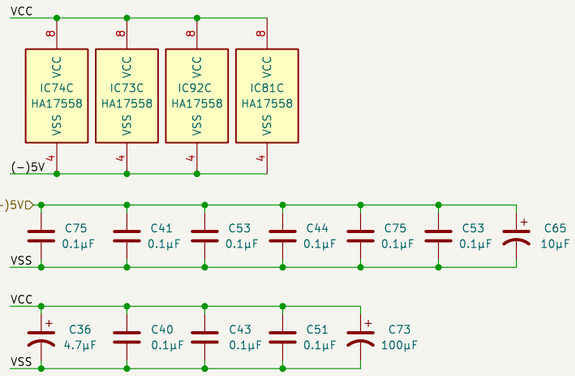
JOTEGO

Sheet: /top/char/char pixels/
File: chpixel.kicad_sch

Title:

Size: A3	Date:	**Rev:**
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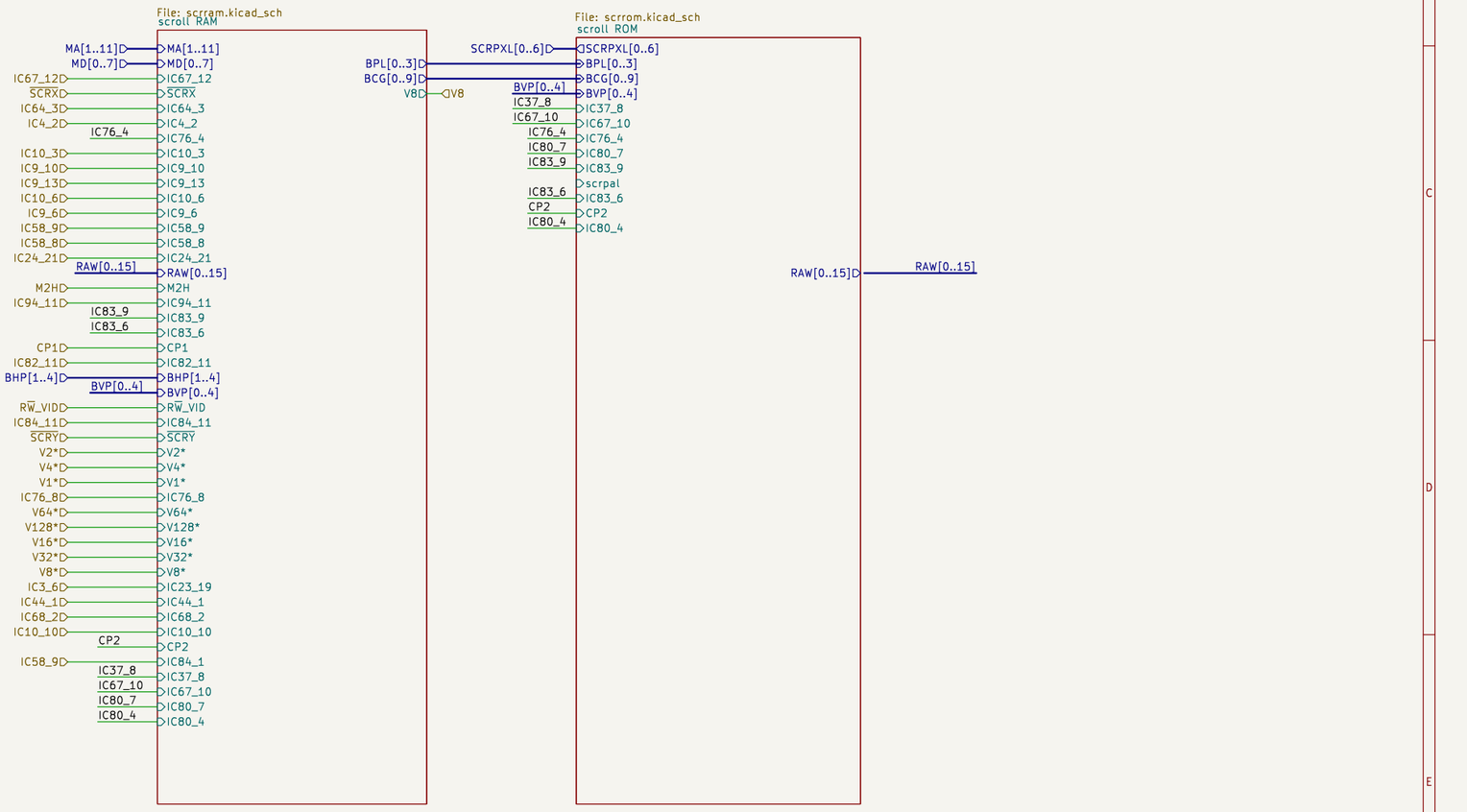


JOTEGO

Sheet: /top/sound/analog/
File: analog.kicad_sch

Title:

Size: A3	Date:	**Rev:**
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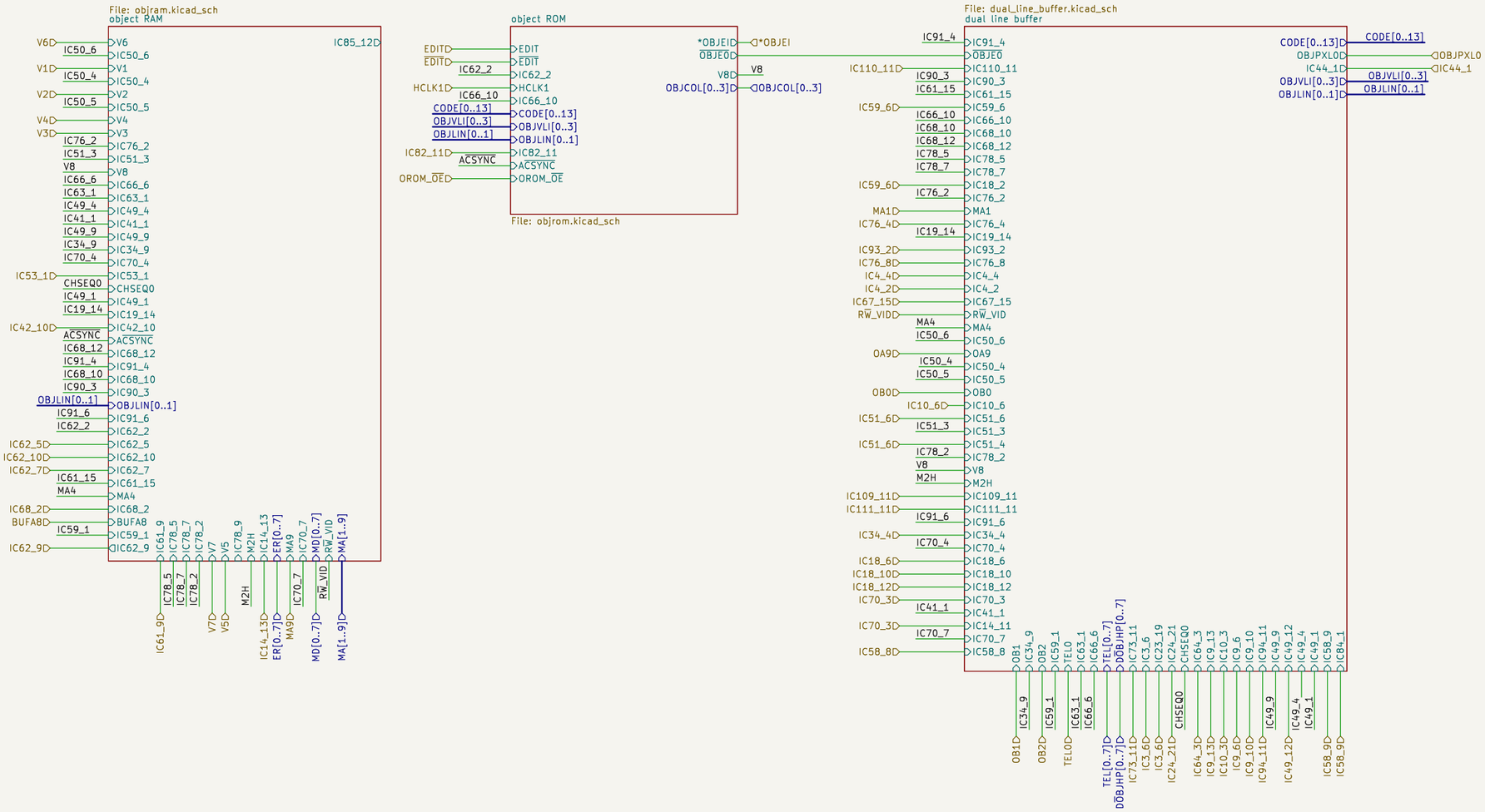


JOTEGO

Sheet: /bottom/scroll/
File: scroll.kicad_sch

Title:

Size: A3	Date:	**Rev:**
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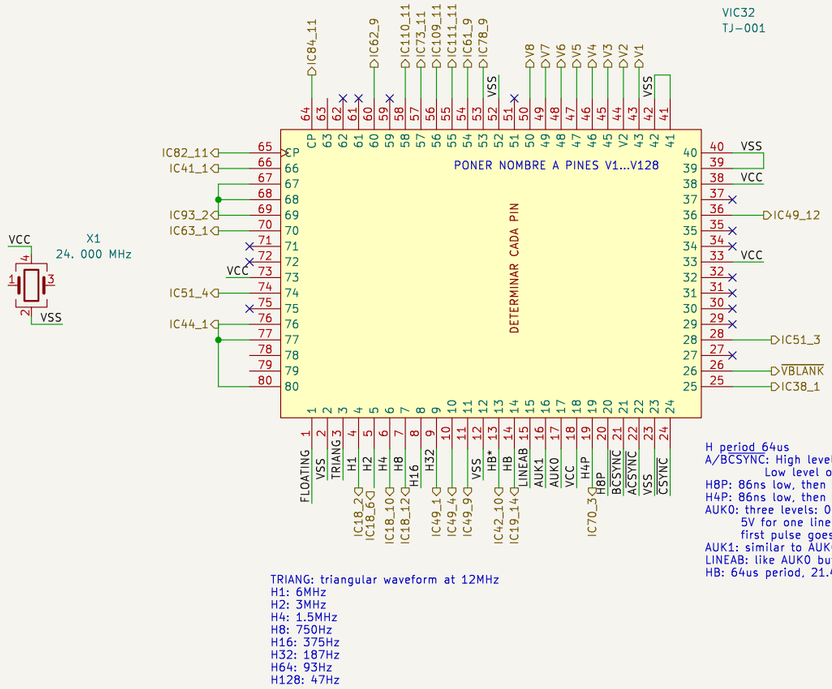
JOTEGO

Sheet: /bottom/object/

File: object.kicad_sch

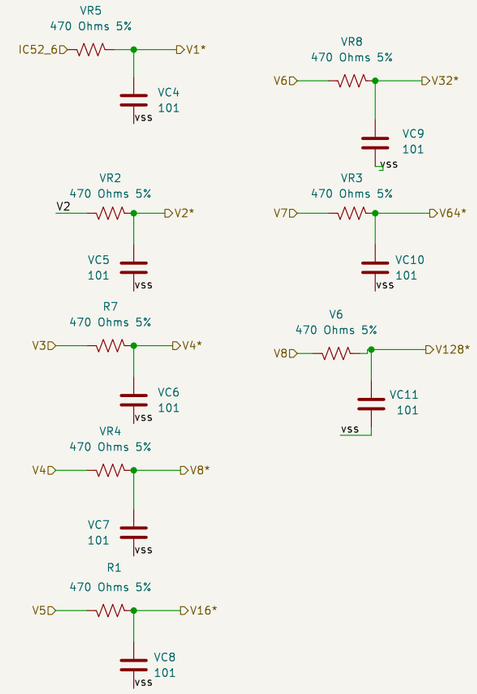
Title: Objects

Size: A3	Date:	Rev:
KiCad E.D.A. 8.0.5		Id: 13/20

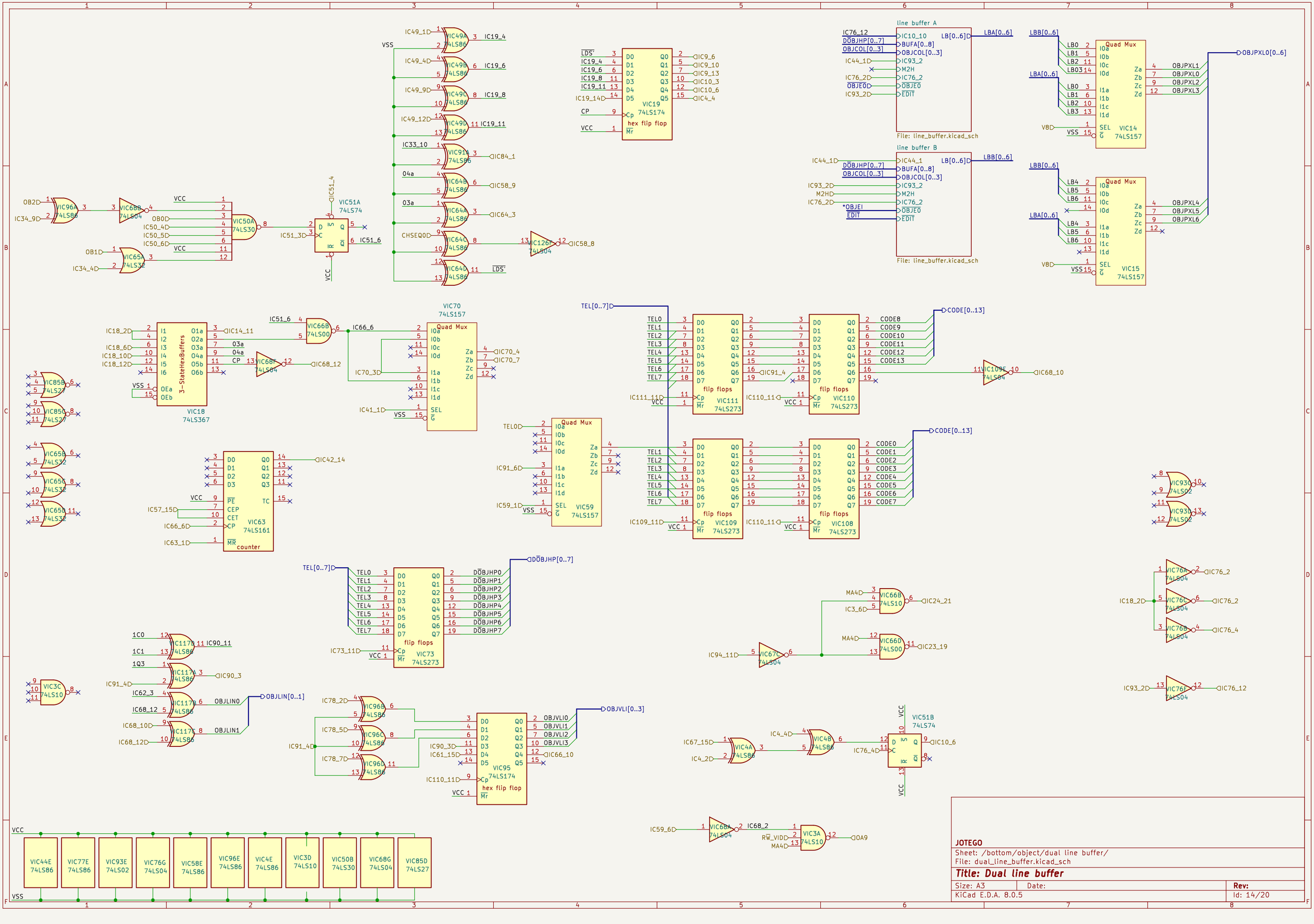


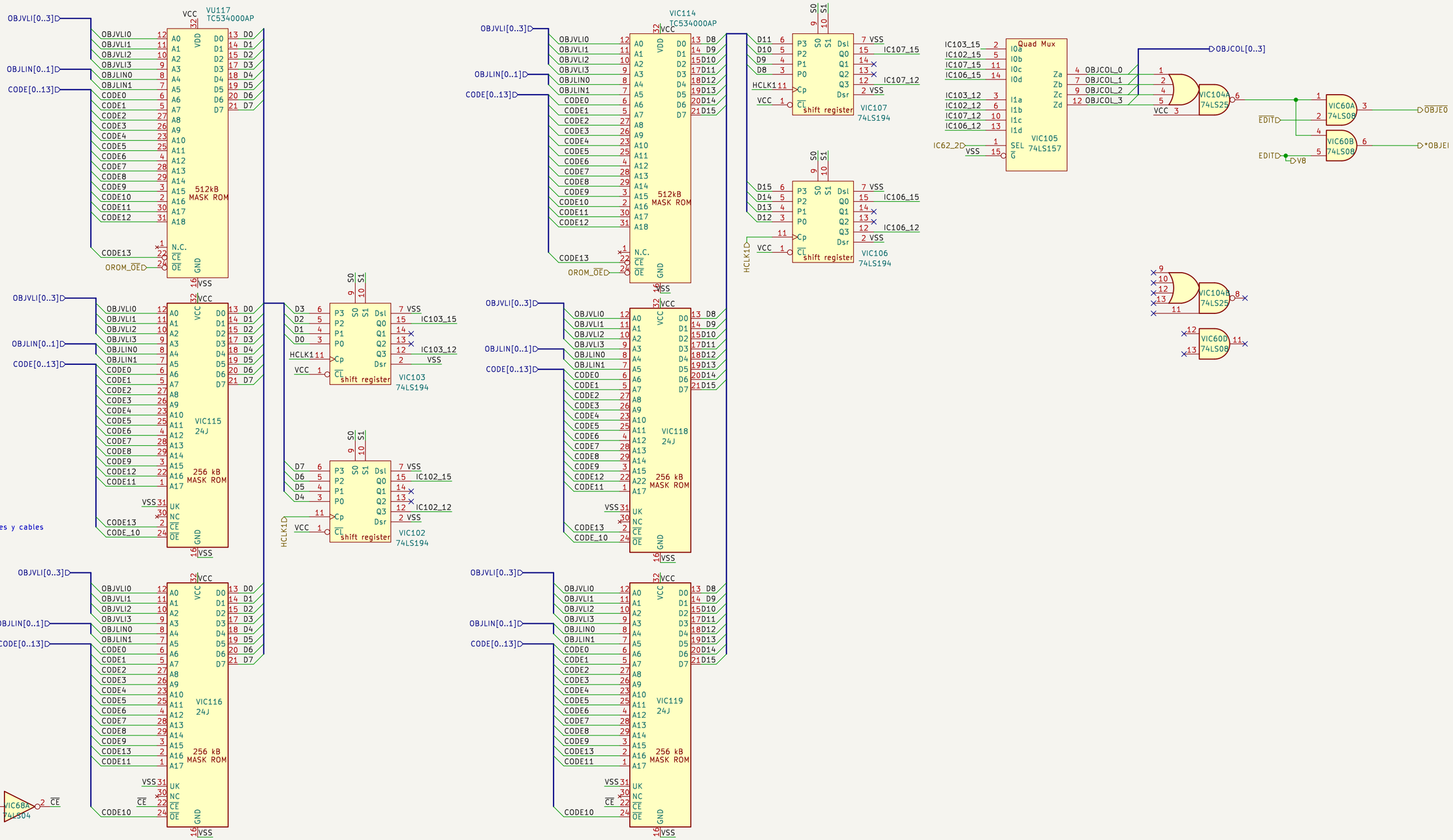
TRIANG: triangular waveform at 12MHz
H1: 6MHz
H2: 3MHz
H4: 1.5MHz
H8: 750Hz
H16: 375Hz
H32: 187Hz
H64: 93Hz
H128: 47Hz

H period 64us
A/BCSYNC: High level on visible area 5V, during VBLANK ~2V
Low level on visible area ~2V, during VBLANK 0V
H8P: 86ns low, then high. Period 668ns
H4P: 86ns low, then high. Period 332ns
AUK0: three levels: 0, 2 and 5V. subperiod 2.66us. period 128us
5V for one line, then pulses down during the next line. The first pulse goes 5->0V, the rest 5->2V
AUK1: similar to AUK0
LINEAB: like AUK0 but purely digital (0/5V)
HB: 64us period, 21.4us high



JOTEGO		
Sheet: /bottom/timing/		
File: timing.kicad_sch		
Title:		
Size: A3	Date:	Rev:
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line buffer A
File: line_buffer.kicad_sch
line buffer B
File: line_buffer.kicad_sch
Quad Mux
flip flops
hex flip flop
3-StateHexBuffers
counter
JOTEGO
Sheet: /bottom/object/dual line buffer/
File: dual_line_buffer.kicad_sch
Title: Dual line buffer
Size: A3
Date:
Rev:
KiCad E.D.A. 8.0.5
Id: 14/20

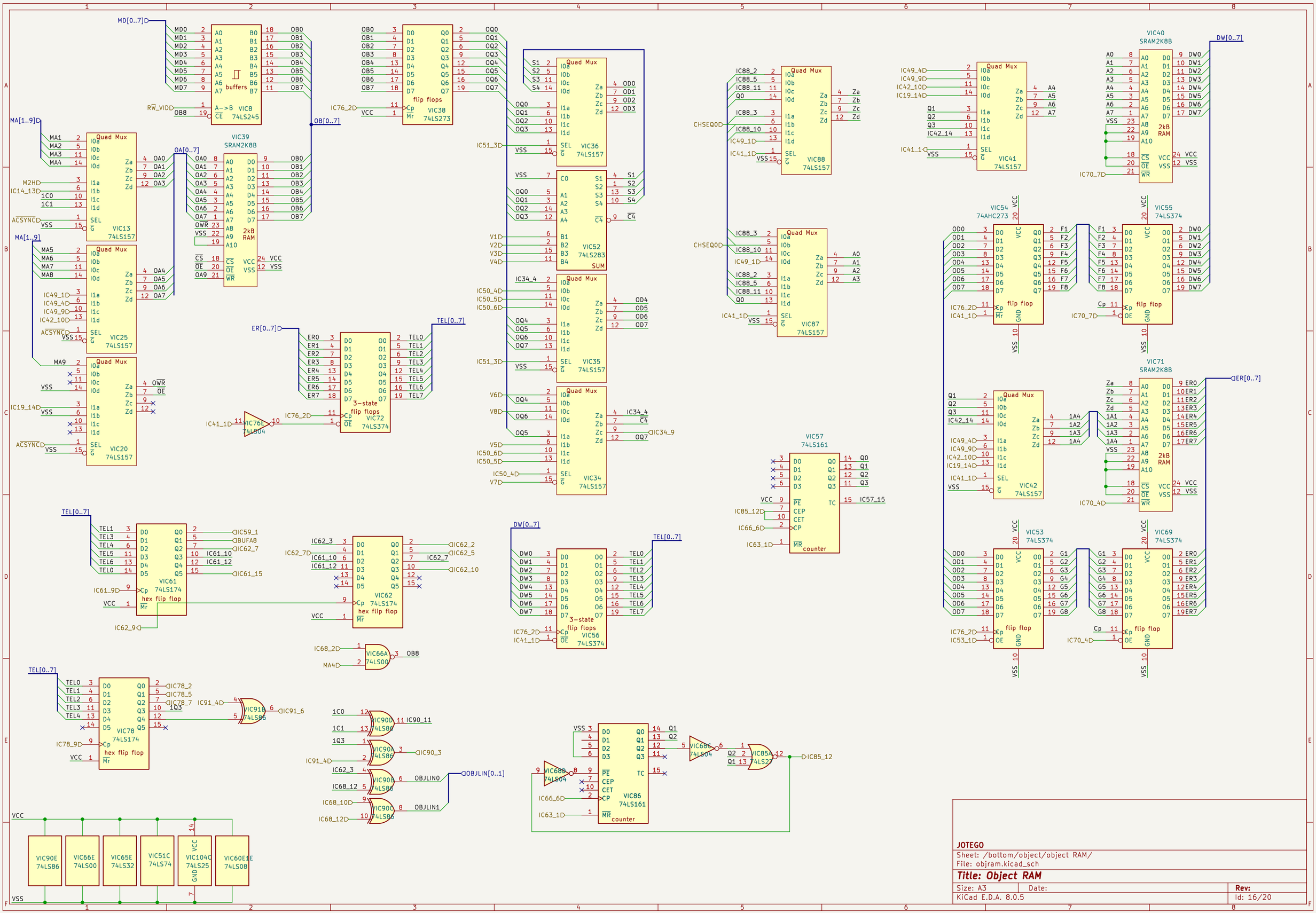


JOTEGO

Sheet: /bottom/object/object ROM/
File: objrom.kicad_sch

Title: Object ROM

Size: A3	Date:	Rev:
KiCad E.D.A. 8.0.5		Id: 15/20

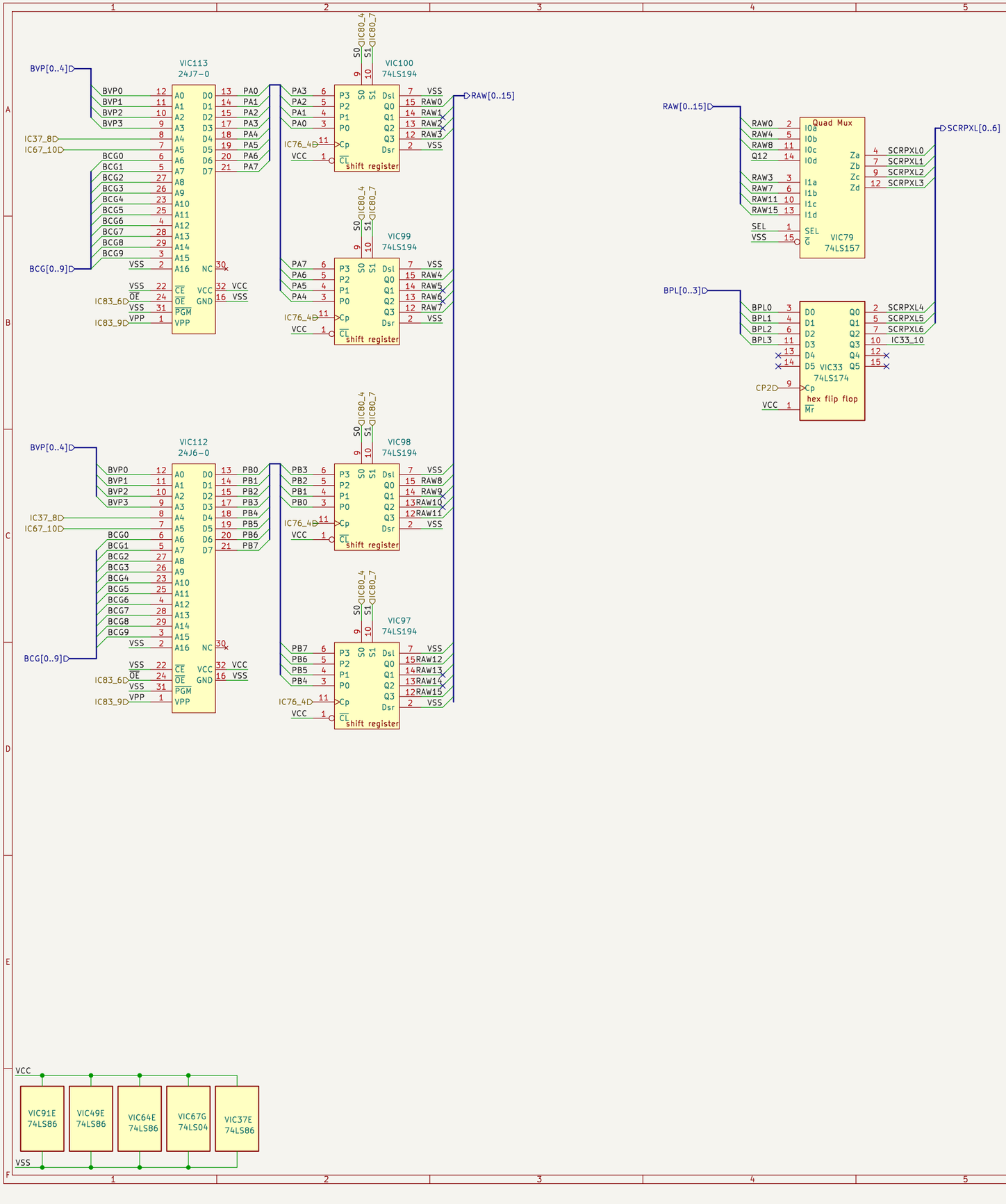


JOTEGO

Sheet: /bottom/object/object RAM/
File: objram.kicad_sch

Title: Object RAM

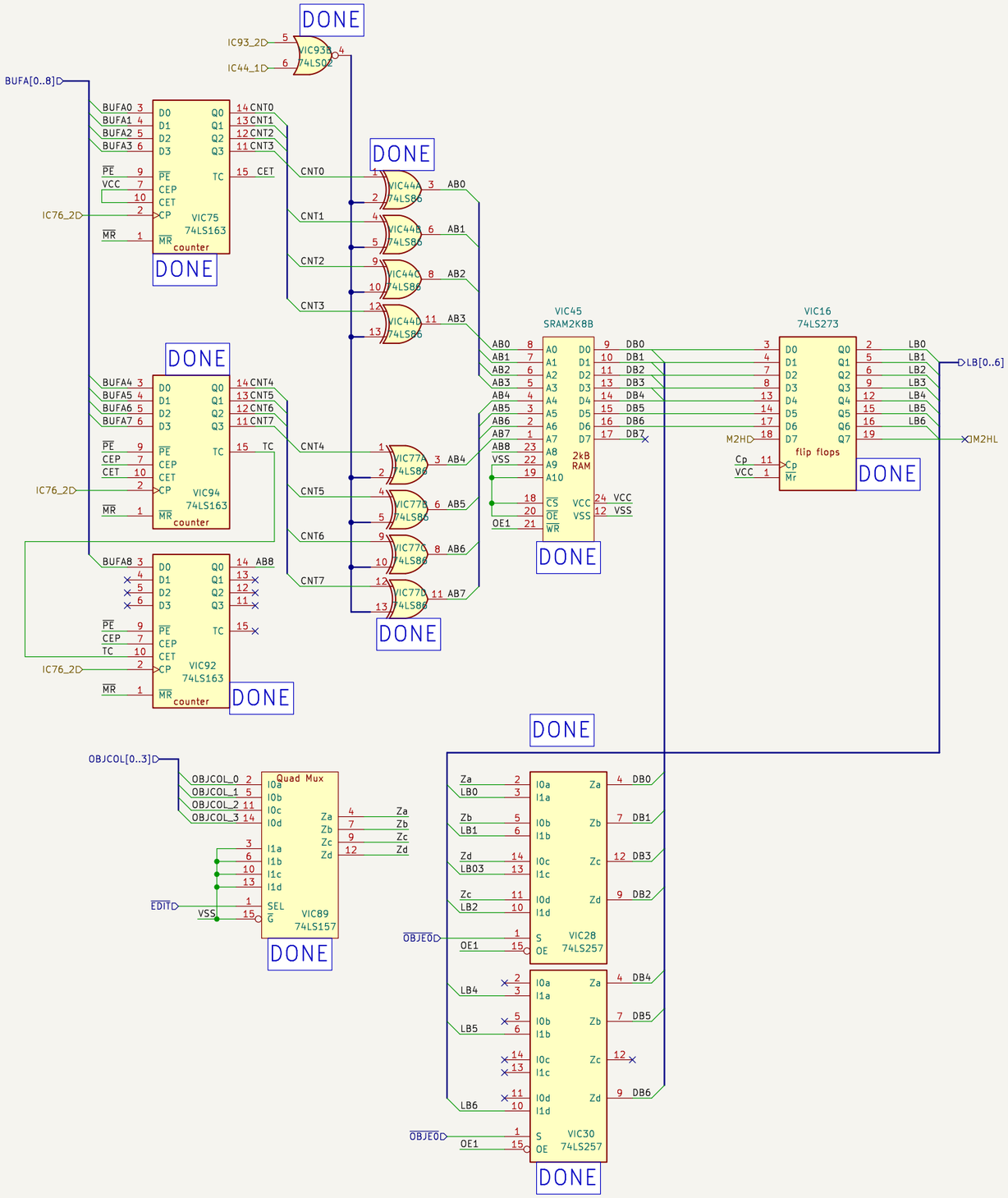
Size: A3	Date:	Rev:
KiCad E.D.A. 8.0.5		Id: 16/20



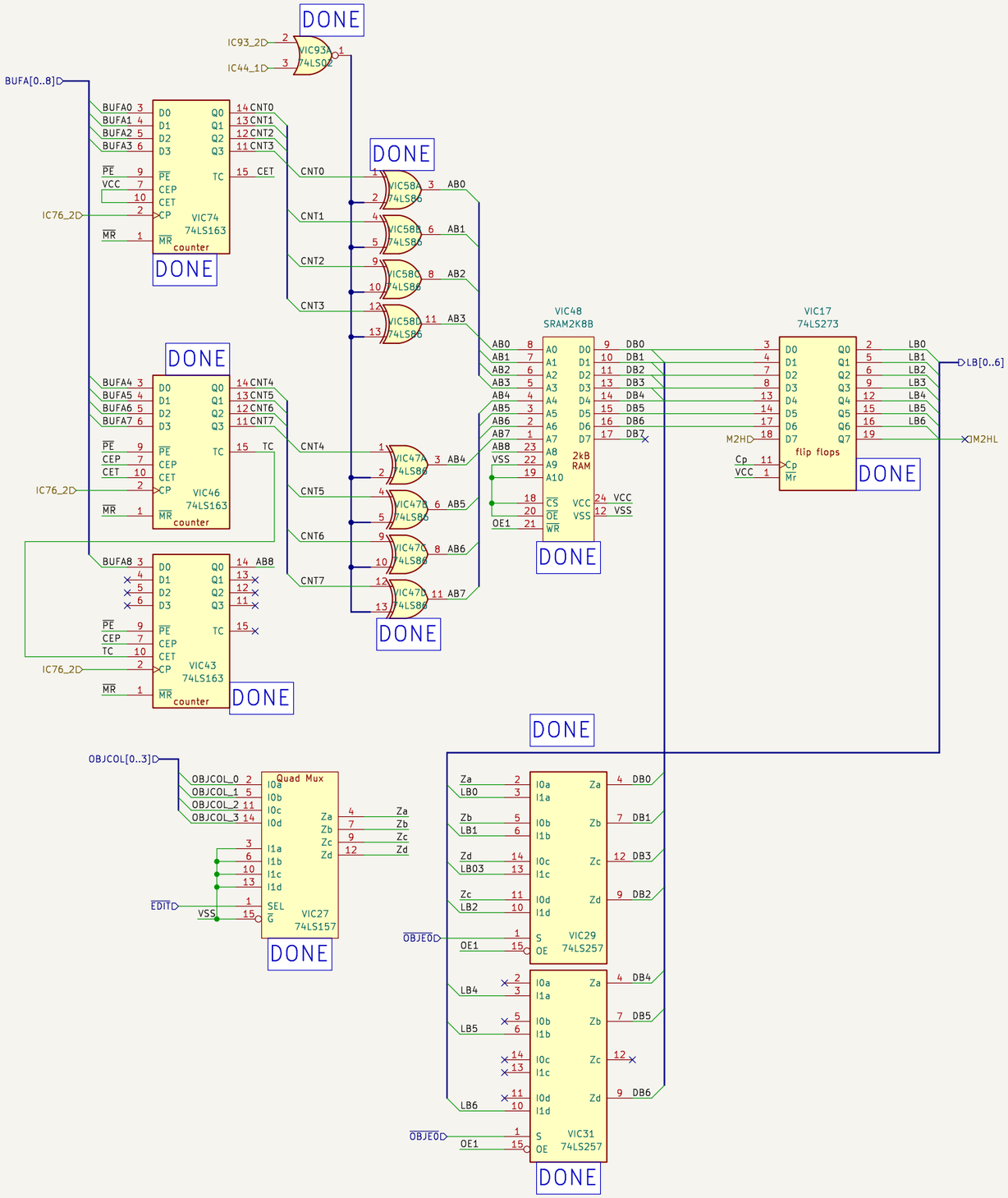
JOTEGO	
Sheet: /bottom/scroll/scroll ROM/	
File: scrrom.kicad_sch	
Title:	
Size: A3	Date:
KiCad E.D.A. 8.0.5	Rev: Id: 17/20

MD[0..7]
MD0 MD1 MD7 MD6 MD5 MD4 MD3 MD2
SCRX
VCC
flip flops
VIC9 74LS273
VSS
C0 A1 A2 A3 A4 B1 B2 B3 B4 C4
S1 S2 S3 S4
VIC7 74LS283 SUM
IC64_3D IC58_9D IC58_8D IC9_6D
Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8
VIC21 74LS283 SUM
IC9_10D IC9_13D IC10_3D IC10_6D
IC4_2
VC12 VC13 VC14 VC15 VC16 C17 VC18 VC19
VIC5 74LS273
BHP[1..4]
BHP0 BHP1 BHP2 BHP3 BHP4 BHP5 BHP6 BHP7
C76_4D
MA[1..11]
M2HD
IC94_11
Quad Mux
VIC12 74LS157
VIC26 74LS157
VIC10 74LS157
IC26_6 IC26_10 IC26_13
IC10_3 IC10_6D ABC
RW_VIDD
VIC24 SRAM2K8B
2kB RAM
IC24_21
BVP[0..4]
BVP0 BVP1 BVP2 BVP3
VIC6 74LS283
DVPAS0 DVPAS1 DVPAS2 DVPAS3
SA1 AR2 AR3 AR4
VIC11 74LS273
SCRY
C0
VIC22 74LS283
DVPAS4 DVPAS5 DVPAS6 DVPAS7
DVPAS[0..7]
VIC4C 74LS86
IC67_12
BCG[0..9]
BCG0 BCG1 BCG2 BCG3 BCG4 BCG5 BCG6 BCG7 BCG8 BCG9
IC82_11
VIC82 74LS273
VIC81 74LS273
CP1
IC84_11
VIC84 74LS273
VIC83 74LS273
BPL0 BPL1 BPL2 BPL3
BPL[0..3]
IC83_6 IC83_9
MD[0..7]
RW_VIDD
IC23_19D
buffers
A->B
VIC23 74LS245
V1* V2* V4* V8* V16* V32* V64* V128*
IC76_8D
VIC1 74LS273
VIC2 74LS273
IC19_14D
VIC76D 74LS04
VIC3C 74LS86
VIC3D 74LS86
VIC37B 74LS86
VIC37C 74LS86
IC37_8
VIC67E 74LS04
IC67_10
VIC37A 74LS86
IC44_1D
V8
VIC4D 74LS86
VIC3A 74LS86
VIC3B 74LS86
VIC37B 74LS86
IC68_2D
IC10_10D
VIC60C 74LS08
VIC67D 74LS04
CP2
VIC80 74LS157
IC80_4 IC80_7
IC84_1D
VIC91D 74LS86
VIC91C 74LS86

JOTEGO
Sheet: /bottom/scroll/scroll RAM/
File: scram.kicad_sch
Title: Scroll RAM
Size: A3 Date: Rev:
KiCad E.D.A. 8.0.5 Id: 18/20



Sheet: /bottom/object/dual line buffer/line buffer A/		
File: line_buffer.kicad_sch		
Title: Line Buffer		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 8.0.5		Id: 20/20



Sheet: /bottom/object/dual line buffer/line buffer B/
File: line_buffer.kicad_sch

Title: Line Buffer

Size: A3	Date:	**Rev:**
KiCad E.D.A. 8.0.5		Id: 21/20